Skills Summary

Reading a Conic Equation Without Losing a Sign or a Square Root

Use this reference while completing your worksheet or when studying.

Skill 1 — Centre and radius of a circle

Standard form: $(x-h)^2 + (y-k)^2 = r^2$, centre (h, k), radius r.

- The form *subtracts* the centre, so read h and k by setting each binomial to zero. A printed $+$ gives a negative coordinate.
- The right-hand side is r^2. The radius is its square root.

Example: Find the centre and radius of $(x-2)^2 + (y+5)^2 = 81$.
Step 1. The two things asked for are the centre and the radius, and both are hidden by an operation — a subtraction in the binomials and a square on the right — so undo one at a time.
Step 2. Set each binomial to zero: $x - 2 = 0$ and $y + 5 = 0$, so the centre is $\mathbf{(2,\ -5)}$.
Step 3. The right side is r^2, so take its square root: $r = \mathbf{9}$.

Try it: Find the centre and radius of $(x+6)^2 + (y-1)^2 = 25$.

check: centre $\mathbf{(-6,\ 1)}$, radius $\mathbf{5}$

(!) Watch out: $(x+6)$ does *not* mean the centre is at 6. Ask what value of x makes the bracket zero — that is the coordinate.

Skill 2 — Reading an ellipse's denominators

Standard form: $\dfrac{(x-h)^2}{a^2} + \dfrac{(y-k)^2}{b^2} = 1.$

- Each denominator is a *square*: the semi-axis is its square root.
- The *larger* denominator marks the major axis, whichever term it sits under — not the term written first.
- Full axis length = twice the semi-axis.

Example: Give both semi-axes of $\dfrac{x^2}{36} + \dfrac{y^2}{16} = 1$.
Step 1. Semi-axis lengths are asked for and the denominators are squares, so this is a square-root reading, not a copy.
Step 2. Under x^2: the semi-axis is $\sqrt{36} = \mathbf{6}$, and since 36 is the larger denominator this is the semi-*major* axis, lying horizontally.
Step 3. Under y^2: the semi-minor axis is $\sqrt{16} = \mathbf{4}$.

Try it: Give the semi-major and semi-minor axes of $\dfrac{x^2}{9} + \dfrac{y^2}{64} = 1$, and say which way the major axis runs.

check: semi-major **8** (vertical), semi-minor **3**

Skill 3 — Which way a hyperbola opens

Standard forms: $\dfrac{(x-h)^2}{a^2} - \dfrac{(y-k)^2}{b^2} = 1$ opens left and right; $\dfrac{(y-k)^2}{b^2} - \dfrac{(x-h)^2}{a^2} = 1$ opens up and down.

- The two squared terms are *subtracted*. The branches open along the variable whose term is **positive**.
- The vertex distance is the square root of the *positive* term's denominator. The subtracted term's denominator never gives a vertex.

Example: Which way does $\dfrac{x^2}{25} - \dfrac{y^2}{4} = 1$ open, and where are its vertices?

Step 1. The x term carries the plus sign, so the branches open left and right; the vertices then sit on the x-axis and are found by setting $y = 0$.

Step 2. With $y = 0$ the equation becomes $\dfrac{x^2}{25} = 1$, so $x^2 = 25$ and $x = \mathbf{-5}$ or $x = \mathbf{5}$: the vertices are five units either side of the centre.

Try it: Which way does $\dfrac{y^2}{49} - \dfrac{x^2}{9} = 1$ open, and where are its vertices?

check: opens up and down; $y = \mathbf{-7}$ or $y = \mathbf{7}$

(!) Watch out: Reaching for the denominator under the *subtracted* term puts the vertices where the curve has no points at all. Find the plus sign first, every time.